AKASH RAO

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EDUCATION

Auburn University, Auburn, Alabama, USA *January 2025 – Present*
PhD in Computer Science

Auburn University, Auburn, Alabama, USA *August 2023 – August 2025*
MS Data Science and Engineering

Visvesvaraya Technological University, Belagavi, Karnataka, India *August 2017 – August 2021*
BTech Computer Science

PUBLICATIONS

1. **"FSP-DETR: Few-shot Prototypical Parasitic Ova Detection,"** Shubham Trehan, Udhav Ramachandran, Akash Rao, Ruth Scimeca, and S.N. Aakur *(Co-First Author, Published, IEEE/CVF WACV 2026)*
https://openaccess.thecvf.com/content/WACV2026/html/Trehan_FSP-DETR_FewShot_Prototypical_Parasitic_Ova_Detection_WACV_2026_paper.html

Diagnosing parasitic infections from microscope images is hard when only a handful of labeled examples exist per parasite. Built FSP-DETR, a single object detection model that handles few-shot detection of known parasites, open-set recognition of unfamiliar ones, and zero-shot generalization to entirely new parasite types from just a few examples.

- Paired a class-agnostic DETR detection backbone with prototype-based classification to improve few-shot detection accuracy over the best existing method with only 5 labeled examples per parasite (mAP 0.066 vs. 0.039), outperforming eight competing object detection systems including YOLO and DETR.
- Added a background-prototype rejection mechanism for open-set recognition, correctly flagging 3 out of 4 parasite species never seen during training (74% open-set recall) with zero labeled examples of them, making the model viable for catching newly emerging infections.
- Trained the embedding space with an energy-based objective (prototype matching + KL-divergence distillation + contrastive InfoNCE-style alignment) that generalized to two unrelated few-shot medical imaging tasks, ranking first on both blood cell and malaria detection using just 5 examples per class.
- Enabled zero-shot recognition of new parasite types by averaging a few support-image embeddings into a class prototype at inference, adapting to new classes with no retraining or code changes.
- Curated and publicly released a benchmark of 3,001 annotated images across 20 parasite species with seen/unseen splits, now a standard testbed for evaluating few-shot and open-set detection methods.

2. **"HyperTaxa: Multiscale Genome Representations for Generalized Microbial Taxonomic Inference,"** Akash Rao, Sushim Gupta, Vishalini Ramnath, Akhilesh Ramachandran, and S.N. Aakur *(First Author, Accepted, IEEE BIBM 2026)*

As DNA sequencing produces a flood of new organisms, identifying them becomes hard when a species is rare, poorly studied, or missing from reference databases entirely, exactly the case for emerging pathogens. HyperTaxa tackles this by learning genome representations that can place a bacterial species into the tree of life even with no training examples of it, supporting zero-shot (ZSL), generalized zero-shot (GZSL), and few-shot inference.

- Encoded genomes as multiscale hyperspectral k-mer transition fields aligned to a taxonomy-aware embedding, achieving up to 75% order-level and 66% family-level zero-shot classification accuracy on unseen bacterial species (97-species benchmark), far exceeding prior pseudo-image and transformer-graph baselines.
- Designed a two-stage Node2Vec-to-MDS embedding alignment that preserves genus-family-order hierarchy, reaching 64–70% generalized zero-shot (GZSL) accuracy at higher taxonomic ranks on a 1,110-species training / 219 unseen-species evaluation while reducing seen-class bias.
- Applied few-shot prototype inference to a 100-pathogen clinical diagnostic panel, retrieving the correct pathogen in 7 of 9 real raw-read DNA samples and producing compact candidate sets for downstream BLAST confirmation, with no retraining.
- Built a cascaded genus-to-species inference strategy that narrows candidates by genus before species prediction to boost fine-grained accuracy, and showed a lightweight CNN encoder outperformed larger ResNet and Vision Transformer (ViT) backbones.
- Formulated the task as open-set representation learning with a regression-based alignment objective (chosen over contrastive learning for sparse, imbalanced genome data), enabling inference over species with no training examples.

3. **"Class Geometry as Supervision for Sample-Efficient Open-World Detection,"** Akash Rao, Zhou Chen, Revanth Reddy Palem, Udhav Ramachandran, Ruth Scimeca, and S.N. Aakur *(First Author, Under Review, AAAI 2027)*
https://arxiv.org/abs/2608.12698

Open-world detectors must recognize known objects, reject unfamiliar ones, and absorb new categories over time, but prototype-based detectors learn each class as an isolated anchor and ignore how classes relate. Class-Geometry Supervision (CGS) treats those relationships as a training signal, constraining the learned feature space to preserve real visual and semantic dissimilarities between classes for sample-efficient open-world, few-shot, and open-set detection.

- Introduced a dissimilarity-preserving training objective that aligns the learned class-prototype distances with a target class-geometry matrix (built from visual morphology and language-derived class descriptions), added on top of the standard detection loss without changing the detector architecture.
- Applied CGS to novel-class insertion to place new categories into an existing detector by their visual/semantic relationships rather than as isolated anchors, improving novel-class detection accuracy roughly 2.5x (mAP 0.063 to 0.156) and recall from 0.26 to 0.37.
- Used the geometry-organized prototype space to sharpen open-set rejection of unknown objects, raising open-set accuracy 36% (mAP 0.045 to 0.061), and improved few-shot biomedical detection (5-shot mAP 0.066 to 0.076) while preserving known-class performance.
- Transferred CGS to the standard COCO open-world benchmark (OW-DETR) under frozen-head and full fine-tuning setups, increasing unknown-object recall across multiple task stages with a controllable precision/recall trade-off for deployment tuning.
- Proved formal guarantees (a prototype-graph distortion bound and neighborhood-order preservation) establishing when the method keeps class relationships intact, and ran ablations isolating visual class-geometry as the strongest supervisory signal over text-derived and random variants.

WORK EXPERIENCE

Graduate Teaching Assistant, Auburn University
Fall 2026 – Present

- Serve as Teaching Assistant for CPSC 3700: Software Engineering I under Rodrigo Sardinas and CPSC 4970: Special Topics in Artificial Intelligence & Machine Learning under Dr. Shaffer.
- Support course delivery through grading, holding office hours, and assisting students with assignments and core concepts in software engineering, AI, and machine learning.

Graduate Research Assistant, ICAMS
October 2024 – Fall 2026

- Solely built a fully offline, private camera surveillance and streaming system (React, MQTT, Docker, Flask, Raspberry Pi) with motion detection and real-time video prioritization, currently deployed across two lab floors to monitor student activity with no cloud dependency.
- Developed a local-first machine-telemetry data-collection platform (Next.js, FastAPI, PostgreSQL, MQTT, Docker) that routes live sensor data from configurable CNC and manufacturing machines into session-based datasets, plus a Redis-backed dashboard surfacing real-time machine metrics such as temperature and usage.
- Led development of an MRP (Material Requirements Planning) platform (Next.js, FastAPI, PostgreSQL, Redis) to manage manufacturing workflows, inventory, and scheduling, featuring JWT-based role access control (employee/supervisor/admin), bulk CSV/XLSX import pipelines, and Alembic-managed migrations; an ongoing project that continues in active development.
- Built a lab job-management application (React Native, Flask, PostgreSQL) integrated with the Paperless Parts API, enabling supervisors to create and assign incoming job orders while students view, track, and complete them.
- Collaborated on an RFID-based lab attendance system (Arduino, React) that tracks student login/logout times on the lab floor via an RFID scanner.
- Served as technical lead and mentor for a team of undergraduate developers as the sole PhD student on the team, ensuring code quality, architecture consistency, and timely delivery across all lab software projects.

Software Engineer, Dcode Solutions L.L.C.
October 2021 – June 2023

- Built an end-to-end asset tracking system for the U.A.E. Ministry of Monitor and Control (Android/Java, MSSQL, React) with a three-tier approval workflow, letting warehouse staff tag asset locations via Honeywell barcode scanners while managers and supervisors reviewed, approved, and audited checkouts through a web dashboard.
- Engineered the system to operate fully offline for warehouses without internet, syncing to the server post-operation, and added GPS tracking for outdoor assets and RFID support alongside the primary barcode workflow.
- Served as solo technical lead integrating a Bartender- and Excel-based stocktaking and inventory solution into Dubai Duty-Free's existing systems, meeting client-specific requirements and constraints.
- Delivered and customized a stock take application for multiple major U.A.E. retailers (Sharaf DG, Koi, and others), analyzing each client's business processes and providing training to support adoption.

- Collaborated closely with a small team of senior engineers, gaining hands-on full-stack and mobile development experience across mission-critical government and enterprise deployments.

Software Engineer Intern, Scientia Innovations *October 2020 – December 2020*

- Independently developed a cross-platform marketplace application for online courses and study material using Flutter, handling the full build and third-party API integrations solo while receiving mentorship from a senior engineer.
- Adopted Flutter as an early-stage framework during its initial rise, delivering a functional cross-platform app while still an undergraduate student.

PROJECTS

Automated Image Collection with Robotic Arm

- Developed a Python control pipeline for a Dobot robotic arm to autonomously reposition a target object and capture images from multiple angles and viewpoints.
- Automated the dataset collection process for building custom object-recognition datasets, reducing manual data-gathering effort.

Advanced Video Analysis via DETR

- Built a video object detection pipeline in Python using Detection Transformers (DETR).
- Applied the transformer-based detector to video datasets to extract spatial and temporal features across frames.

SKILLS AND CERTIFICATIONS

Programming Languages: Python, Java, JavaScript, SQL

ML / AI: PyTorch, TensorFlow, Scikit-learn, Gensim; local LLM tools (Ollama, AnythingLLM, Llama 3.2)

Web & Backend: React, Next.js, Node.js, FastAPI, Flask, REST APIs

Mobile: React Native, Flutter, Native Android

Databases & Infrastructure: PostgreSQL, MSSQL, Redis, Docker, MQTT

Hardware & Embedded: Raspberry Pi, Arduino, Dobot Robotic Arm, barcode/RFID systems

Certifications: Machine Learning (Stanford University, Coursera), Industry 4.0, Docker for Developers